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Risa 3D

Analysis of a Stable Structural Frame

## Analysis of a Stable Structural Frame

### Procedure:

Complete a stable structure in Risa 3D. Give the output of the structure and compare the results to my handwritten calculations.

### Results:

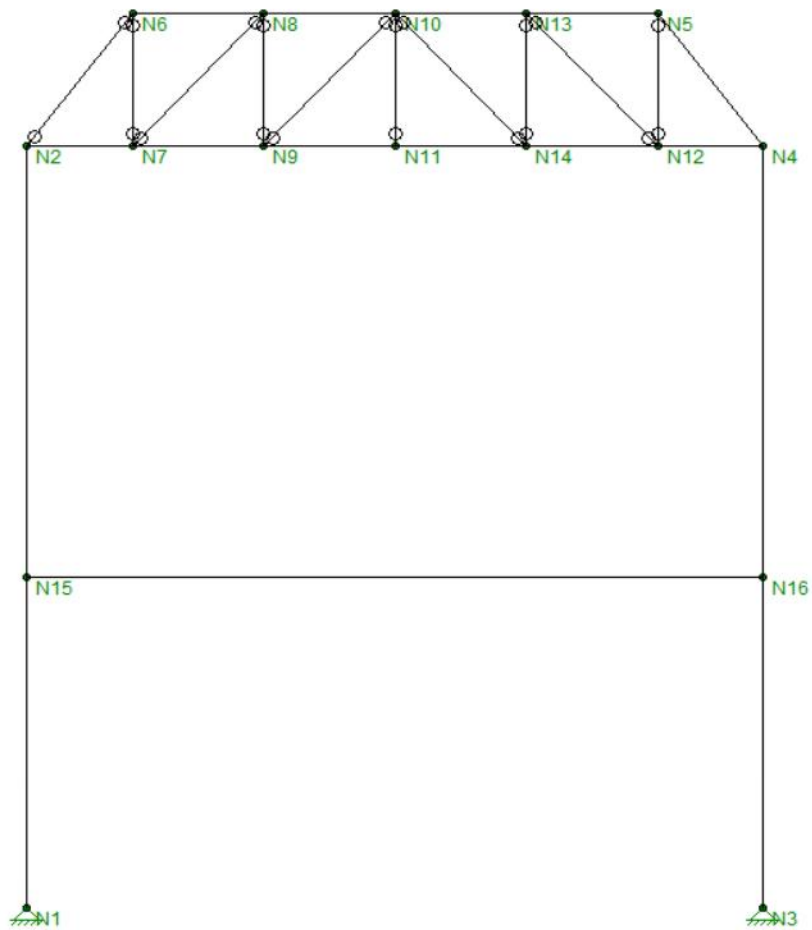
Tables 1 and 2 provide the joint coordinate and member data to complete my structural model. Figure 1 is the result of the combination of data in Table 1 and Table 2.

| 3D Member Primary Data                                                                                                         |       |         |         |         |           |               |        |              |           |             |
|--------------------------------------------------------------------------------------------------------------------------------|-------|---------|---------|---------|-----------|---------------|--------|--------------|-----------|-------------|
| Primary   Advanced   Hot Rolled   Cold Formed   Wood   Concrete Beam   Concrete Column   Aluminum   Stainless   RISAConnection |       |         |         |         |           |               |        |              |           |             |
|                                                                                                                                | Label | I Joint | J Joint | K Joint | Rotate... | Section/Sh... | Type   | Design List  | Material  | Design ...  |
| 1                                                                                                                              | M1    | N1      | N2      |         |           | CRECT20X20    | Column | Rectangular  | Conc4000  | Col-Rules   |
| 2                                                                                                                              | M2    | N3      | N4      |         |           | CRECT20X20    | Column | Rectangular  | Conc4000  | Col-Rules   |
| 3                                                                                                                              | M3    | N2      | N4      |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 4                                                                                                                              | M4    | N4      | N5      |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 5                                                                                                                              | M5    | N5      | N6      |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 6                                                                                                                              | M6    | N6      | N2      |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 7                                                                                                                              | M7    | N6      | N7      |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 8                                                                                                                              | M8    | N7      | N8      |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 9                                                                                                                              | M9    | N8      | N9      |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 10                                                                                                                             | M10   | N9      | N10     |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 11                                                                                                                             | M11   | N10     | N11     |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 12                                                                                                                             | M12   | N5      | N12     |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 13                                                                                                                             | M13   | N12     | N13     |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 14                                                                                                                             | M14   | N13     | N14     |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 15                                                                                                                             | M15   | N14     | N10     |         |           | Chord         | Beam   | Double Angle | A36 Gr.36 | Truss Rules |
| 16                                                                                                                             | M16   | N15     | N16     |         |           | CRECT24X20    | Beam   | Rectangular  | Conc4000  | Col-Rules   |

**Table 1:** Includes data for each member

| Joint Coordinates and Temperatures |       |        |        |        |          |                          |
|------------------------------------|-------|--------|--------|--------|----------|--------------------------|
|                                    | Label | X [ft] | Y [ft] | Z [ft] | Temp [F] | Detach...                |
| 1                                  | N1    | 0      | 0      | 0      | 0        | <input type="checkbox"/> |
| 2                                  | N2    | 0      | 18.5   | 0      | 0        | <input type="checkbox"/> |
| 3                                  | N3    | 21     | 0      | 0      | 0        | <input type="checkbox"/> |
| 4                                  | N4    | 21     | 18.5   | 0      | 0        | <input type="checkbox"/> |
| 5                                  | N5    | 18     | 22.5   | 0      | 0        | <input type="checkbox"/> |
| 6                                  | N6    | 3      | 22.5   | 0      | 0        | <input type="checkbox"/> |
| 7                                  | N7    | 3      | 18.5   | 0      | 0        | <input type="checkbox"/> |
| 8                                  | N8    | 6.75   | 22.5   | 0      | 0        | <input type="checkbox"/> |
| 9                                  | N9    | 6.75   | 18.5   | 0      | 0        | <input type="checkbox"/> |
| 10                                 | N10   | 10.5   | 22.5   | 0      | 0        | <input type="checkbox"/> |
| 11                                 | N11   | 10.5   | 18.5   | 0      | 0        | <input type="checkbox"/> |
| 12                                 | N12   | 18     | 18.5   | 0      | 0        | <input type="checkbox"/> |
| 13                                 | N13   | 14.25  | 22.5   | 0      | 0        | <input type="checkbox"/> |
| 14                                 | N14   | 14.25  | 18.5   | 0      | 0        | <input type="checkbox"/> |
| 15                                 | N15   | 0      | 10     | 0      | 0        | <input type="checkbox"/> |
| 16                                 | N16   | 21     | 10     | 0      | 0        | <input type="checkbox"/> |

**Table 2:** Includes the exact joint coordinates in order to create Figure 1



**Figure 1:** The model that is the result of tables 1 and 2 data

### Conclusion:

I was successful in creating this model as there weren't any discrepancies in regards to data input. Which thereby helped me to create a stable model in Risa 3D that also matched my handwritten calculations. Stability and safety are very important as a structural engineer. My goal was achieved.